

LiquidAI / LFM2.5-Encoder-230M

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LFM2.5-Encoder-230M

LFM2.5-Encoder is a family of multilingual bidirectional encoders built on the LFM2 architecture, available in two sizes:

- **LFM2.5-Encoder-230M** (*this model*) — a lightweight encoder for tight latency and memory budgets, punching above its size class.
- **LFM2.5-Encoder-350M** — a larger sibling for maximum downstream quality.

Both are masked language models with full bidirectional attention, designed to be fine-tuned into task-specific models (classification, token classification, retrieval, reranking, and semantic

Downloads last month
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Safetensors

Model size 0.2B params

Tensor type F32 Chat template

Files info

Inference Providers NEW

Fill-Mask

This model isn't deployed by any Inference Provider.

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Model tree for LiquidAI/LFM2.5-E...

Base model LiquidAI/LFM2.5-230M-Ba...

Finetuned (7) this model

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 **Liquid Nanos** Collection

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similarity) across 15 languages, and to run efficiently on-device.

Find more details about our encoders in our [blog post](#).

Key highlights:

- **Highly capable for its size.** On par with the best similarly sized encoders and well ahead of our own retrieval siblings.
- **General-purpose.** 8k context, strong across NLI, paraphrase, sentiment, and multilingual tasks.
- **Fast and on-device.** Matches or beats ModernBERT throughput, with a long-context edge on CPU; runs in the browser on WebGPU.

 **Demos:** We built the demos below from fine-tuned LFM2.5-Encoders. Each one runs in a CPU-only Hugging Face space:

- [Zero-shot prompt routing](#) — define your own routing lanes as free text. The model scores the whole prompt against every lane in one pass.
- [Zero-shot policy linting](#) — check text against your company's rules, written as free text. It scores every token against every rule in one pass.
- [Spell checking](#) — correct misspellings token by token.
- [PII detection](#) — spot and remove 40 kinds of personal information across 16 languages.
- [Masked-diffusion text generation](#) — bonus: run the encoder as a chatbot that generates text

 **Article mentioning** LiquidAI/LFM2....

LFM2.5-Encoders for Fast Long-Con...

 LiquidAI • 2 days ago •  50

by iteratively unmasking instead of left to right.

 Model details

Property	LFM2.5-Encoder-230M	LFM2.5-Encoder-350M
Type	Bidirectional encoder (masked language model)	Bidirectional encoder (masked language model)
Backbone	LFM2	LFM2
Total parameters	~229.7M	~354.5M
Hidden size	1024	1024
Vocabulary size	65,536	65,536
Context length	8,192 tokens	8,192 tokens
License	LFM Open License v1.0	LFM Open License v1.0

Supported languages: English, German, Spanish, French, Italian, Dutch, Polish, Portuguese, Arabic, Hindi, Japanese, Russian, Turkish, Vietnamese, Chinese (15).

Architecture. LFM2.5-Encoder is built on the LFM2 hybrid backbone, which interleaves gated short-convolution blocks with grouped-query attention. For encoder use, the causal mask is replaced with full bidirectional (non-causal) attention and the

model is trained with a masked language modeling head. The encoder body is exposed as `Lfm2BidirectionalModel`; masked-LM loading uses `Lfm2BidirectionalForMaskedLM`. Both are wired through `auto_map` and require `trust_remote_code=True`.

```
Lfm2BidirectionalForMaskedLM(  
    (lfm2): Lfm2BidirectionalModel  
    (lm_head): Linear(in_features=1024, out_fe  
)
```

Training. LFM2.5-Encoder-230M is adapted from the LFM2 base and trained with a masked language modeling objective on a large multilingual corpus. Pre-training uses a two-stage schedule that extends the context window to up to 8,192 tokens.

We recommend fine-tuning LFM2.5-Encoder-230M for a range of downstream tasks, such as:

- **Text classification:** sentiment, topic, intent/routing, moderation, and business-text linting.
- **Token classification:** named-entity recognition, span extraction, and sequence labeling.
- **Retrieval and reranking:** a backbone for dense embedding or late-interaction (ColBERT-style) retrievers.
- **Semantic similarity:** STS, paraphrase, and duplicate detection.
- **Natural language inference and extractive QA:** sentence-pair reasoning and answer-span extraction.

Its small footprint makes it especially well-suited to on-device and browser (WebGPU) deployment.

How to run

Install the latest version of transformers:

```
pip install -U transformers
```

Run masked-token prediction:

```
from transformers import AutoModelForMaskedLM
import torch

tok = AutoTokenizer.from_pretrained("LiquidAI/LiquidAI-2.5-Encoder-230M")
mlm = AutoModelForMaskedLM.from_pretrained("LiquidAI/LiquidAI-2.5-Encoder-230M")

text = f"The capital of France is {tok.mask_token}"
enc = tok(text, return_tensors="pt")
with torch.no_grad():
    logits = mlm(**enc).logits
pos = (enc["input_ids"][0] == tok.mask_token).nonzero().squeeze(-1)
print([tok.decode([t]).strip() for t in logits[pos]])
# -> ['Paris', 'Strasbourg', 'Paris', 'Lyon']
```

For downstream tasks, load the encoder body and attach your own head (classification, token classification, regression, retrieval):

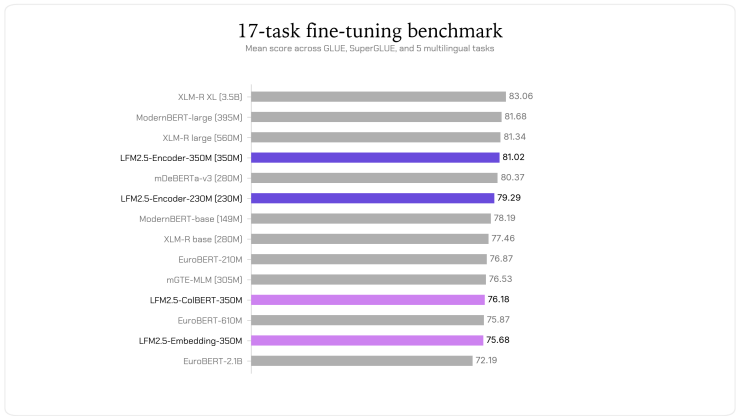
```
from transformers import AutoModel
body = AutoModel.from_pretrained("LiquidAI/LiquidAI-2.5-Encoder-230M")
```

If your GPU supports it, we recommend using LFM2.5-Encoder-230M with Flash Attention 2 to reach the highest efficiency. To do so, install Flash Attention as follows, then use the model as normal:

```
pip install flash-attn
```


Performance

For each benchmark task, we run a full supervised fine-tune and report that fine-tuned model's score. The results below span 14 models across 17 tasks from GLUE, SuperGLUE, and multilingual classification tasks. The full evaluation harness is open-sourced in the [eurobert-repro](#) repository.



17-task results (avg@5 fresh seeds ± std)

Rank	Model	Params	17-task mean	± std
1	XLNet-R XL (3.5B)	3.5B	83.06	±1.16
2	ModernBERT-large (395M)	395M	81.68	±2.49
3	XLNet-R large (560M)	560M	81.34	±1.66
4	LFM2.5-Encoder-350M (ours)	350M	81.02	±1.00
5	mDeBERTa-v3 (280M)	280M	80.37	±1.06

Rank	Model	Params	17-task mean	$\pm$ std
6	LFM2.5-Encoder-230M (ours)	230M	79.29	± 1.02
7	ModernBERT-base (149M)	149M	78.19	± 1.39
8	XLM-R base (280M)	280M	77.46	± 1.63
9	EuroBERT-210M	210M	76.87	± 2.00
10	mGTE-MLM (305M)	305M	76.53	± 1.85
11	LFM2.5-ColBERT-350M	350M	76.18	± 1.25
12	EuroBERT-610M	610M	75.87	± 2.03
13	LFM2.5-Embedding-350M	350M	75.68	± 0.83
14	EuroBERT-2.1B	2.1B	72.19	± 5.59

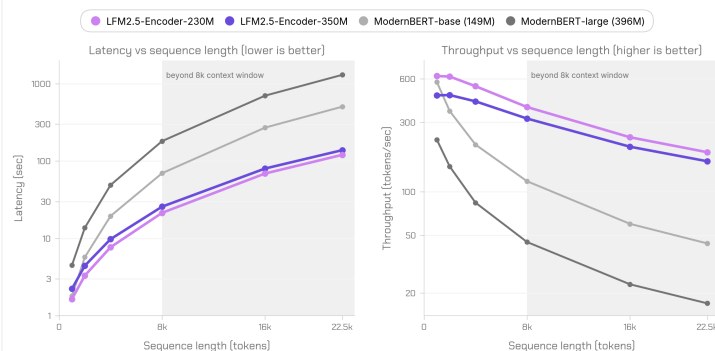
► Click to expand per-task results — all 17 tasks
(avg@5 fresh seeds $\pm$ std)

Inference speed

The LFM2 backbone was built for fast inference, and the encoders inherit it. While ModernBERT-base is faster at short sequences in Apple GPU inputs, LFM2.5-Encoders overtake it as inputs grow. At long input sequences of 8k on CPU, the encoders run 3.3× faster than ModernBERT-base.

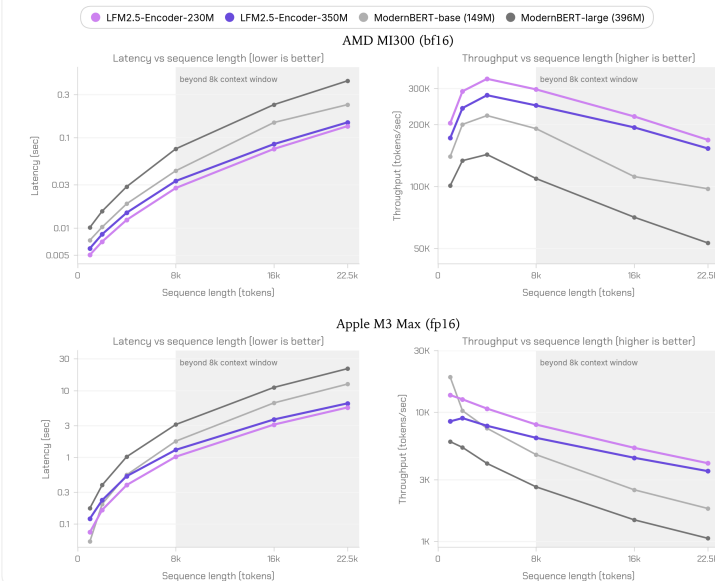
Inference latency & throughput

Apple M3 Max (fp16), batch 1, median



Inference latency & throughput

Batch 1, median



Fine-tuning

LFM2.5-Encoder-230M follows standard BERT-style fine-tuning. Attach a task head to the encoder body and train end-to-end. Suggested starting points (tune per task):

Hyperparameter	Suggested range
Learning rate	1e-5 – 5e-5
Warmup ratio	0.1
Weight decay	0.1
Epochs	3 – 20 (early stopping, patience 3)

Hyperparameter	Suggested range
Precision	bf16 autocast (fp32 master weights)

Contact

- Got questions or want to connect? [Join our Discord community](#).
- If you are interested in custom solutions with edge deployment, please contact [our sales team](#).

Citation

```
@article{liquidAI2026Encoders,
  author = {Liquid AI},
  title = {LFM2.5-Encoders: Fast at Long Con
  journal = {Liquid AI Blog},
  year = {2026},
  note = {www.liquid.ai/blog/lfm2-5-encoders},
}
```